

SPI-M-O Committee Lessons Learnt

FINAL

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SPI-M-O COMMITTEE: LESSONS LEARNT

PURPOSE OF DOCUMENT

1. This document sets out lessons learnt gathered from SPI-M-O committee members and observers relating to the role that the committee played during the recent swine flu pandemic. The document is intended to be an internal DH document with the aim of being informative for future pre-pandemic planning.
2. The lessons learnt have been drawn together through consulting individually with members and some observers of the SPI-M-O committee, see Appendix A for a full list of those interviewed. Interviewees were asked to provide feedback on those aspects that were important to them (rather than going through a list of all potential aspects) and were also asked to give their top 3 lessons learnt, since this allowed for some prioritisation of those lessons that were most important to the committee as a whole (see page 4).
3. The lessons learnt in this report focus on the 'process' aspects of the SPI-M-O committee rather than the scientific or epidemiological aspects. However, the report does cover some scientific aspects where science and process intertwine. Epidemiologists on the committee will consider more fully the scientific and epidemiological lessons learnt from the pandemic in due course.
4. In some areas the lessons learnt are clear-cut. However, the document also seeks to highlight those areas where the lessons are less well defined or where views of members or observers diverge. As such, the lessons learnt in these areas could benefit from further consideration or clarification.
5. Since the interviews were concerned with lessons learnt, they necessarily focused on those aspects that could benefit from improvement, however, where positive feedback was provided this has been included.

BACKGROUND

6. During non-pandemic times, the Scientific Pandemic Influenza committee (SPI) and its sub-group on modelling (SPI-M) generally met quarterly. At the outset of the epidemic, as was planned, SPI transformed into the Scientific Advisory Group for Emergencies (SAGE) and SPI-M took on the role of SPI-M-O (the operational sub-group on modelling).
7. The SPI-M-O committee met weekly, with a few exceptions, over the course of the pandemic, from 30th April until 11th December 2009.
8. The severity of the pandemic has emerged to be mild to moderate compared to some previous outbreaks. Any lessons learnt should therefore be considered in that context.

PRIORITIES FOR LESSONS LEARNT

Interviewees were asked to prioritise their lessons learnt and this was used to determine the top priorities of the committee as a whole. These are as follows:

Process

- **Managing expectations** of senior officials of what epidemiological modelling and analysis can and can't provide (see page 26)
- **Better links with the Chief Medical Officer's (CMO) office** (see page 19)
- **Planning assumptions**, review content of and process for production and promulgation (see page 16)
- **Data availability and accessibility**, review process for collection of and access to data relevant for epidemiological modelling and analysis (see page 19)

Science / Epidemiology

- **Serology**, review whether seroepidemiological studies can be better planned ahead of time (see page 24)
- *Epidemiologists on the committee to further consider the scientific lessons learnt in due course*

SPI-M-O PROCESS: GENERAL

9. It was felt, by most that commented, that the SPI-M-O process had worked generally well.

SPI-M-O COMMITTEE'S REMIT

Initial transition period

10. It was commented by a few interviewees that there was initial uncertainty as to the role of the committee at the outset of the epidemic as SPI transformed into SAGE and SPI-M into SPI-M-O. It was also commented that, as it became apparent that 'Real-Time Modelling' (RTM) wasn't to happen as planned (and the remit of the committee seemed to change to a more general group modelling effort) the revised remit could have benefited from further clarification.

Operational versus advisory

11. There was some feeling that it was not always clear whether SPI-M-O was fulfilling an operational or advisory role. It was commented by one interviewee that, whilst SPI-M-O's remit was to support SAGE and CCC operationally (i.e. by providing regular consensus and interpretive statements) it also needed to continue with its advisory role in terms of supporting DH policy, JCVI and ECDC directly (i.e. not through SAGE). It was felt that, during the epidemic, there was pressure for SPI-M-O's role to be narrowed to just its operational role in supporting SAGE but that both its operational and advisory roles were vital and needed to be properly recognised.

4 nations

12. There was strong feedback from some that the '4 nations' coordination over the course of the epidemic (not just in terms of SPI-M-O's role) could have been considerably improved. In terms of SPI-M-O's role, it was unclear to some whether the committee's remit was or should have been to focus on English data or the four nations as a whole. It was also commented by one interviewee that it was a long time before a Scottish representative attended the committee meetings and that there were no representatives from Wales or Northern Ireland. However, it was highlighted that the differences in surveillance schemes, policies and population sizes for each of the four nations (for example, Scotland is approximately the size of one English SHA) should be taken into account when considering the extent of SPI-M-O's role across the four nations.

More 'arms-length'?

13. It was considered by some, particularly in light of the difficulties experienced in conveying SPI-M-O's consensus via the planning assumptions (see page 16) whether it is feasible for the committee's function to be made more 'arms-length'. This could allow the committee to become less political and act more scientifically thereby avoiding significant delays with its statements. However, it was acknowledged that there is need for government involvement and confidentiality in some stages of the process and as such it was unclear how this would work.

SPI-M-O and DH

14. A few interviewees commented that it was not always clear how SPI-M-O related to DH. For example, when SAGE made a request for analysis it was not always clear if the request was directed at DH or SPI-M-O. It was also felt by a few that, at a senior official level, DH could have better used and understood its committees.

SPI-M-O COMMITTEE PROCESS AND ADMINISTRATION

Make-up of Committee

15. The SPI-M-O committee was essentially made-up of epidemiologists from the HPA, an epidemiologist from Imperial College providing an “official opposition” or “challenge” role, and independent epidemiologists from the London School of Hygiene and Tropical Medicine and the University of Warwick. DH civil servants provided the secretariat function for the committee, with observers from DH policy, DH analytical groups, Government Office for Science and Cabinet Office regularly attending the committee meetings.
16. Those that commented on the make-up of the committee felt that its composition worked generally well with comments ranging from “fine” to “very good”.
17. It was felt by some interviewees that having epidemiologists from a variety of institutions on the committee was very important in ensuring that the committee’s debate was intellectually robust and that having a variety of different models to compare was powerful and reassuring to policy makers. It was also commented that it had been important that the core of the committee had been composed purely of epidemiological modellers which had allowed them to focus on the epidemiological aspects of the task. One interviewee commented that the make-up of the group was extremely effective and should be supported and not changed.
18. However, two observers commented that it would have been beneficial to have a statistician on the committee (as a member) since it was felt that, at times, the statistical aspects of the analysis were not considered as much as they might have been.
19. It was also commented that the committee could have benefitted from greater links with the HPA, for example, in enabling the committee to have greater input into the design of epidemiological surveys etc.
20. It was also commented, however, that although the make-up of the committee was fine, the process for selection of members to the committee had been a little opaque.

Chairing

21. During the initial period of the epidemic (April to July 2009) the committee was chaired by Dr Peter Grove (DH analytical lead on pandemic influenza). However, as this initial period came to a close it was felt that the committee would benefit from an external chair and responsibility for chairing was transferred to the independent epidemiologists on the committee (i.e. from the University of Warwick and the LSHTM). However, in practice, due to non-availability at times, the chairing switched between DH and the independent epidemiologists.

22. Those that commented on the chairing of the committee felt that it was generally “OK” to “good” and one interviewee commented that the style of chairing meant that the committee was very responsive to the needs of SAGE. However, some interviewees did raise a question as to whether, ideally, the chair should be an external scientist (i.e. not DH or involved in research into the epidemic).
23. One interviewee commented that, in practice, it had been beneficial that three members were able to chair the meetings. However, another interviewee felt that having a consistent chair throughout the process was a better approach in terms of flow of information and thinking between meetings. It was also suggested that the chair should also be a member of SAGE, however, this could pose a problem if a flexible approach to chairing was deemed to be the best approach (e.g. 3 members able to chair) since all may not be members of SAGE.

Independent peer review of SPI-M-O’s output

24. It was suggested (via a similar ‘lessons learnt’ review of the SAGE committee’s role during the pandemic) that SPI-M-O’s output should be peer reviewed by an epidemiological expert on SAGE. However, SPI-M-O members generally felt that, whilst independent peer review outside of SPI-M-O might theoretically be the best approach, it was unclear how this would work in practice. Most SPI-M-O meetings over the course of the epidemic overran and many of those involved in the process were short on time and resource and therefore questions were posed as to the feasibility of introducing another layer in the SPI-M-O and SAGE process.

Members’ roles and responsibilities

25. Due to SPI-M-O and SAGE requirements, a number of interviewees felt that there was little time to conduct other research or to publish academic papers on the epidemic. One interviewee commented that the time commitment required by the process, particularly for independent epidemiologists had been considerable.
26. Some interviewees also posed the question of whether, during the most intensive periods of the epidemic, it was possible to quality assure the work of other members and felt that further clarification in terms of members’ responsibilities for quality assurance would be beneficial.

Making use of committee members’ expertise

27. A number of interviewees commented that the scientific discussions during the committee meetings were “OK” to “excellent” and that the free-style of the scientific discussions at the start of each meeting was a good structure. However, this contrasted with views from a few that, at times, there was not

enough debate or disagreement. One interviewee suggested that this might have been because the committee had not been required to deal with any particularly marginal situations (e.g. a particularly difficult trade-off between certain policies). Another interviewee suggested that there might be more formal ways of structuring the debate and getting views from the experts and that this could benefit from further consideration.

Deputising

28. It was unclear to a few interviewees what the official procedure was for deputising. One interviewee also felt that the distinction between deputies and members and, in particular, their roles in finalising the Consensus and Interpretive statements could benefit from further clarification.

Minutes and action points

29. The committee secretariat did not officially record minutes during committee meetings (it is not an official requirement of a sub-committee to do so). However, action points during the meetings were recorded.
30. Those that commented generally felt that not recording minutes was the correct approach given the time and resource constraints. It was felt that producing and agreeing minutes would have been very burdensome and that the official papers produced by the committee (particularly the Consensus and Interpretive Statements) were sufficient in terms of producing an audit trail. However, one interviewee highlighted that, although not an issue during this pandemic, the non-recording of minutes meant that there was no means for members to express dissent.
31. One interviewee also commented that recording minutes (and making them public) could make committee members overly cautious in their conclusions or recommendations.

Frequency of meetings

32. It was generally felt that the frequency of the committee meetings was about right. However, a few commented that that it would have been possible to meet less frequently, whilst one interviewee felt that the meetings had to be held once a week by default in order to fit in with the wider process.
33. It was suggested by one interviewee, that with better management of expectations (see page 26), it might have been possible to insist that the committee had less frequent meetings at certain points during the epidemic. For example, the group might only convene when there was sufficient data for consideration with potential for significant interpretations to be made from it.

34. It was highlighted by some that, had the epidemic been more severe and required more frequent meetings, this could have posed resource or capacity issues for some members.

Role of the secretariat

35. Those that commented felt that the secretariat did an “OK” to “good” job.

Administration of papers

36. It was commented that the administration of the large number of papers for the meetings was an issue at the start of the process, however once additional administrative resource was found the process became manageable.
37. Some interviewees commented that, since papers were sometimes circulated late (often necessarily) this meant that on occasions there was little time for members to assess the work of others and that this posed a question of the extent to which it is the role of members to quality assure the work of others (see paragraph 26). This was particularly an issue for those travelling to the meetings and it was suggested that perhaps travelling members should be given mobile laptop internet access to alleviate this issue to some extent.
38. A number of committee papers, particularly the Consensus and Interpretive Statements were drafted or amended during committee meetings and required ratification by members by the close of play via email. It was commented by some that this created some problems for those travelling to and from the meetings though it was unclear how this could be improved.

Location

39. It was generally felt that the location for the meetings was fine and had to be London.

Teleconference facilities

40. Most that commented on the teleconference facilities felt that it was particularly poor, since it was very difficult to hear the committee’s debate when dialling-in. Some interviewees felt that when dialling-in it was difficult to know when they could contribute to the debate, despite chairs doing a good job of relaying information. It was also noted that, had the epidemic been more severe and the committee had needed to meet briefly every day at critical points, it would not be feasible for the committee to congregate in one locality, therefore highlighting that good remote conferencing facilities are essential.

PRE-PANDEMIC PLANNING

41. It was felt by some interviewees that the large amount of scientifically informed pre-pandemic planning was not made use of as much as it could have been. However, one interviewee commented that the SPI-M modelling summary (containing the pre-pandemic planning assumptions) had been invaluable. It was suggested, by some, that a review of existing pre-pandemic planning in terms of how it fared during the epidemic should be carried out.
42. Some felt that there is a need for a more nuanced approach for dealing with a mild epidemic including how it might be possible, early on, to identify that an epidemic will be mild rather than assuming it will be severe. For example, it was suggested that there could be a hierarchy of pre-pandemic planning documents, for different pandemic severities, with relatively clear-cut criteria as to when each plan should be used and that data from the FF100 database could be used to give some idea of how to switch between different plans. It was also commented by one interviewee that the process for assessing the severity of a pandemic needs to be formally reviewed again.
43. Pre-pandemic planning material included a 15 week single-peaked planning profile. This profile was developed based on the 1918 pandemic (London data). The intention was for this profile to be used in the initial stages of a pandemic and to be replaced by output from RTM. However, in practice, this did not happen and the profile was used as a planning tool over the course of the pandemic (it was used in the July 09 and September 09 versions of the publicly issued planning assumptions and used as part of an Excel planning tool for planners within the NHS). There were mixed views from interviewees as to the usefulness of the profile. Some felt that the profile did fit each of the peaks of the pandemic relatively well, whereas others questioned the use of a single-peaked profile since multiple peaks during a pandemic are likely.
44. Some interviewees also questioned what the best measure for the profile was (i.e. a profile of numbers of clinical cases, GP consultations or deaths etc) and what aspects of the profile were most important to planners (i.e. the peak rate, scale of ramp-up etc). It was generally felt that further consultation with planners on these aspects would be beneficial.
45. It was also generally felt that there was a danger for the profiles to be incorrectly considered as forecasts of the progression of the pandemic and that this was exacerbated by analysis carried out early on in the pandemic to determine whether we were in 'Week 1' of the profile.
46. It was suggested that the pre-pandemic planning profiles could take the form of a number of different profile shapes to convey the potential variation in how a pandemic may progress (for example based on the 3 most recent pandemics) with a watermark to emphasis that the profiles are intended for planning purposes rather than forecasts.

MODELLING & EPIDEMIOLOGY: GENERAL

Mild and prolonged epidemic

47. It was commented by many that there were several difficulties in modelling and interpreting data due to the mild to moderate nature of the disease. In particular, this meant that the distinction between symptomatic and non-symptomatic cases was not well-defined which meant that modelling clinical cases and clinical attack rates was somewhat metaphysical. It was noted that, in a more severe pandemic, where scientific advice would be more critical for decision-making, the severe nature of the pandemic would mean that clearer data would likely be available.
48. It was also noted that the epidemic had been particularly prolonged partly due to the timing of school holidays and, as such, the use of the 15 week profile (used in pre-pandemic planning) and the associated analysis to determine whether we were in 'Week 1' was questioned by one interviewee.

Prescriptive v independent modelling

49. Most that commented felt that independent modelling (as opposed to being prescribed to the modellers by the chair or elsewhere) was generally the best approach since modellers had differing and valid opinions as to the best output measures and that an independent approach allowed experts to focus on aspects that they felt most important, to some degree. However, it was acknowledged that this approach made it more difficult in making the 'mental gymnastics' required to be able to quality assure the work of other members (see paragraph 26) as opposed to a more prescriptive approach to modelling. Some anecdotal evidence also highlighted that an independent approach can lead to issues further downstream (e.g. results shown on different scales, different terminology used etc.) which may need to be dealt with by the secretariat or alternatively require some management of expectations downstream.

Early analysis

50. It was felt that it was natural for the committee to focus on estimating the reproduction number of the disease (R_0) from the FF100 data in the early stages of the epidemic. However, it was commented that, requests early on for very frequent updates of the epidemiological situation (i.e. estimates of R_0 values) led to some over-interpretation of the stochastic elements of the initial stages of the epidemic. The stochasticity pushed R_0 values repeatedly above and below 1, creating potentially unfounded concerns and then relief. It was suggested that, whilst this analysis was important to the SPI-M-O committee, it perhaps should not have been promulgated up to SAGE and CCC until there was a clearer picture. It was also suggested that this issue could be mitigated through managing expectations of SAGE and CCC through the use of a primer (see page 26) to re-iterate that information would

become available in such a way that 'windows of clarity' would arise when informed decisions could be made.

HPA CASE ESTIMATES

Case estimates methodology

51. Throughout the epidemic the HPA has produced a low, mid and high estimate of the number of new clinical cases each week for England. These estimates were reported internally in the HPA weekly situation report and publicly through the CMO's weekly report which headlined the mid estimate (and also quoted the range).
52. Essentially, these weekly estimates were calculated by taking an estimate of the number of GP ILI consultations in England for a particular week (from QSurveillance) and multiplying this figure by the latest estimate of the GP positivity rate (obtained from the RCGP and RMN swabbing schemes) and combining this figure with the number of antiviral authorisations for that week from NPFS (after it became operational on the 23rd July 2009) multiplied by an estimate of the latest NPFS positivity rate (obtained from the NPFS swabbing scheme). This figure was then adjusted to account for the fact that not all clinical cases will seek medical care (through either their GP or NPFS). This adjustment assumed that, before NPFS became operational, between 20% and 50% (central estimate of 29%) of pandemic clinical cases sought medical care with between 30% and 70% (central estimate of 50%) after NPFS became operational. These adjustments therefore enabled a low, mid and high estimate of new clinical cases each week to be calculated. There was also some back-dating of estimates since there was a delay in full swab positivity data becoming available from the various swabbing schemes.

SPI-M-O's views on the HPA case estimates

53. The committee was asked to endorse the HPA case estimates methodology. Whilst a few interviewees commented that producing case estimates is an understandable endeavour, and some felt that some of the criticisms of the methodology had been unfair, the epidemiologists on the committee generally felt that the methodology was not robust and as such did not endorse it. The primary reasons quoted (during the interviews) for the non-endorsement were as follows:
 - There was (and still is) considerable uncertainty as to the proportion of clinical cases that sought medical care (a key parameter in the case estimate methodology) and this proportion was likely to vary considerably over the course of the epidemic.

- For a relatively mild epidemic, as was the case for this epidemic, estimating numbers of clinical cases is somewhat metaphysical because the distinction between symptomatic and non-symptomatic cases was not well-defined
54. There were also some concerns as to the way that these estimates were reported, as follows:
- Only initial estimates were reported in the CMO's report (i.e. it did not ever refer to the revised figures). This had potential to lead to confusion over the true trends in the estimates. It was suggested that it would have been better for the case estimates to have been shown on a graph (showing trends over time) with confidence intervals demonstrating that the latest week's data is based on incomplete information.
55. There were also some concerns relating to how the committee had (or otherwise) been engaged with the decisions made relating to the production of the case estimates:
- Some interviewees questioned why the committee had only been asked to endorse the methodology, devised elsewhere, rather than to devise one itself.
 - One interviewee commented that the committee was asked to consider the methodology too late on in the process (i.e. after estimates using the methodology had already been reported). This meant that it was then difficult to make any changes to the process.
56. It was commented by some that England was unique in the world in producing case estimates and questions were posed as to how other countries were able to resist producing such estimates. It was suggested that, had the frailties in syndromic surveillance been explained better at the outset, there might not have been so much pressure (from outside of the SPI-M-O committee) to produce such estimates (see managing expectations section on page 26).
57. One interviewee also suggested that the way that the case estimates had been calculated, and the data used in the calculations, could have been made more transparent. For example, the data could have been shared on the ImmForm website.

SPI-M-O's suggested alternatives to the HPA case estimates

58. Despite their criticisms of the case estimates methodology, many interviewees acknowledged that there would always be a demand for an indication of the severity and progression of an epidemic and that some sort

of estimates needed to be communicated internally and publicly. It was also suggested by some that if estimates were not produced officially, then the void would likely be filled by others.

59. It was generally felt, however, that providing indicators or trends in ILI and swab positivity etc. was a better option than providing case estimates (due to the reasons set out in paragraph 53) and some suggested, that a figure or graph with no y-axis label, showing the progression of the pandemic (using the trends in ILI and swab positivity) might be a good alternative.
60. It was also identified that in producing indicators there is a trade-off between simplicity and meaningfulness. For example, there will always be pressure for a summary number to be extracted from whatever estimates are provided and that this should perhaps be guarded against.
61. It was generally acknowledged that developing an agreed alternative to the HPA case estimates would require further consideration by the committee.

CONVEYING SPI-M-O'S CONSENSUS

SPI-M-O's Consensus and Interpretive Statements

62. One of SPI-M-O's key functions during the epidemic was to produce a Consensus Statement (CS) and Interpretive Statement (IS) each week. The aim of the CS was to convey the SPI-M-O committee's current evidence base and its latest consensus on the reasonable worst case and anticipated ranges for key epidemiological parameters. The aim of the IS was to review the latest available indicators in order to interpret how the epidemic was progressing. Both these documents were circulated weekly to SAGE members and officials within DH only. (The planning assumptions conveyed relevant aspects of the committee's consensus publicly, but only at discrete points in time - see page 16.)
63. Those interviewees that commented on the process for the production of the weekly CS and IS felt that it was generally good and that a document (or documents) to audit and convey the committee's latest consensus was essential. It was also generally felt that the non-prescriptive format of the statements worked well since this allowed the committee to be flexible about what was important to include in the statements. Some considered whether, based on experiences from this epidemic, it would be worthwhile designing a format for these statements as part of pre-pandemic planning. However, it was felt that this could be problematic since it could be too defined by the characteristics of this epidemic. It was also suggested that having a very prescriptive pre-defined format would raise expectations of its recipients that the committee might not be able to meet.

Issues associated with the Consensus and Interpretive Statements

64. Issues relating to the *content* of the CS and IS were as follows:
- There was feedback that the content of the CS and IS were overly cautious for too long and that there had been a case for drawing in the RWC estimates sooner.
 - One interviewee felt that more could have been done in determining and conveying age dependent Clinical Attack Rates (CAR) and Case Fatality Ratios (CFR). (The CS generally stated estimates for the CAR and CFR for the population as a whole with some commentary on how these rates vary for different age groups.)
 - One interviewee felt that, occasionally, the statistical aspects of the data were not considered enough (see paragraph 18), for example when the Interpretive Statement reported whether swab positivity had increased or decreased. It was, however, acknowledged that, by definition, the Interpretive Statement required some level of judgement regarding interpretation of the data.
65. Issues relating to the *process* for the production of the CS and IS were as follows:
- Some interviewees commented that the distinction between the two documents could have been clearer
 - There was also concern from some as to whether people were reading the CS and IS since large parts of the statement remained the same from week to week and a number of stock phrases were used. It was suggested that, to alleviate this, it might have been beneficial to start the statements from scratch at various points during the epidemic. There were also some questions as to whether the statements were being circulated widely enough and were being taken into account sufficiently in policy making where appropriate.

The Case for Making the SPI-M-O Consensus and Interpretive Statements Public?

66. There was strong concern from some that, at times, there was a big gulf between the committee's latest consensus (documented in the weekly CS and IS) and the information that the NHS and public were receiving through the planning assumptions (which were published only at a few discrete points in time during the epidemic – see page 16). This led to consideration of whether it would be feasible to make the CS and IS public. Some issues associated with this were identified as follows:
- This would require considerable political and communications agreement ahead of time. Alternatively if the SPI-M-O committee's remit was more 'arms-length' (see paragraph 13) this

could allow the committee more freedom in making public statements outside of the political process. For example, the statements could be made public under the proviso that they do not necessarily reflect government policy (which must take account of other factors such as practicality and financial feasibility).

- Making the committee's consensus public on a regular basis could make the committee more cautious in their conclusions and recommendations.

PLANNING ASSUMPTIONS

67. Swine flu planning assumptions were issued in July 2009 for use by the public, NHS and businesses. These assumptions were revised and re-issued in September 2009 and again in October 2009. The planning assumptions were developed collaboratively between DH policy and the SPI-M-O committee, drawing on the latest SPI-M-O Consensus Statement and contained RWC estimates for a number of parameters important to planners e.g. GP consultations, hospitalisations and critical care cases. Prior to their issue, the draft planning assumptions document was also considered and edited for presentation by SAGE, CCS, DH communications and others. Due to this process, there were considerable delays between the SPI-M-O committee agreeing the initial content of the planning assumptions and the assumptions being issued. For example, for the July issue, there was a delay of approximately 6 weeks.
68. There was very strong feedback from interviewees that the formal process for promulgation of the planning assumptions did not work well.
69. The key issues associated with the *process* by which the planning assumptions were issued were as follows:
 - **Resistance to change;** it was felt that the RWC estimates should have been drawn in a lot quicker. Whilst some commented that the committee was somewhat overly-cautious in bringing down the RWC a number of interviewees commented that (outside of the SPI-M-O committee) there was a feeling that the planning assumptions shouldn't change radically. Some anecdotal evidence indicated that officials felt that radically reducing the planning assumptions could breed complacency which could, in the extreme, lead people to stop planning. However, views by many of the interviewees were that bringing the planning assumptions down would mean something more achievable to plan for.
 - **Too many steps in the process;** there was strong opinion that the process for revising the assumptions involved too many steps and was therefore cumbersome and non-transparent.

70. Both these issues led to significant delays in publishing the assumptions, resulting in them being significantly behind the latest epidemiological consensus discussed in the committee meetings (and being conveyed formally, internally, through the consensus and interpretive statements). Linked to this, one interviewee also commented that the planning assumptions did not include a time-stamp and so for those using them it was not possible to determine how recent the evidence was upon which they were based.
71. Interviewees felt that the first issue (reluctance to change) could be ameliorated by managing expectations of officials, through emphasising that the planning assumptions are very likely to be drawn in as further evidence becomes available during the course of an epidemic (though not necessarily). It was suggested that this could involve a primer including examples of the sort of evidence that becomes available over the course of an epidemic and how that evidence would impact the planning assumptions.
72. It was also suggested that reviewing planning assumptions should be made into a formal and regular (monthly, say) streamlined process. It was felt that formalising the process, including obtaining relevant official's agreement to the process ahead of time, along with the management of expectations of how the assumptions will change, could eliminate unnecessary steps in the process that led to delays.
73. It was also suggested that the terms 'planning envelope' or 'planning scenario' might be better than 'planning assumptions' since this would further emphasise that the assumptions are open to being revised downwards as further evidence becomes available.
74. It was also commented that it shouldn't be the role of the SPI-M-O to support the communications strategy of the department and that the committee got too embroiled in this aspect.
75. The key issues associated with the *content* of the PAs were as follows:
- **RWC rates versus numbers**; the first set of planning assumptions, issued in July 2009, gave reasonable worst case estimates for the Clinical Attack Rate and the Case Fatality Ratio. The July 2009 planning assumptions did not go further in explicitly estimating a figure for the number of deaths in the reasonable worst case. However, recipients of the planning assumptions internally and externally multiplied together the reasonable worst case for the Clinical Attack Rate and the upper bound of the Case Fatality Ratio (although the committee warned against this) thereby producing, what some interviewees argued, was an overly-cautious estimate of the number of deaths, and this was incorrectly portrayed as a SPI-M-O forecast.
76. Whilst this was felt to be somewhat a communications issue it was also felt that the structure of the planning assumptions (i.e. the use of reasonable worst case rates for various parameters) was conducive to this sort of issue, in that it was understandable for recipients of the assumptions to multiply the rates together.

77. To avoid this issue, the planning assumptions issued in October 2009 took an alternative approach; whilst still quoting a reasonable worst case estimate for the Clinical Attack Rate they did not give an estimate for the Case Fatality Ratio but instead gave a direct estimate for the number of deaths (and hospitalisations and those in critical care) in the reasonable worst case.
78. It was felt, by those that commented, that using numbers rather than rates for deaths (and hospitalisations etc.) in the planning assumptions was the best approach. However, it was unclear whether pre-pandemic planning assumptions should take this approach or use rates for all parameters (as is currently the case).
- **RWC versus anticipated ranges;** the SPI-M-O Consensus Statements contained both reasonable worst case and anticipated ranges for key epidemiological parameters. However, only the reasonable worst case estimates ever made it into the published planning assumptions.
79. There were strong and divergent views from interviewees as to whether reasonable worst case estimates or anticipated ranges are the best approach for planners.
80. Some argued that if the purpose of the planning assumptions is to specify what people should do then the reasonable worst case is the correct approach but it must therefore be “reasonable”. It was argued that if an anticipated range was then presented alongside this there could be difficulty in keeping planning momentum to the reasonable worst case.
81. Others felt that too much focus on the reasonable worst case meant that planners weren’t hearing the central theme (i.e. what was anticipated rather than a worst case). It was acknowledged that the anticipated range could be wrong but felt that it was still a better approach than planning for a reasonable worst case that might effectively mean planning to send people home or not being able to cope.
82. One interviewee also highlighted that the anticipated ranges (which were stated in the CS but did not make it into the issued planning assumptions) had turned out to be relatively accurate in terms of what actually happened. It was felt that this had effectively meant that, whilst the public were planning to the reasonable worst case, SPI-M-O had also developed relatively accurate estimates of how the pandemic might progress which weren’t made use of publicly.
83. It was suggested that it would be informative to speak to those that made use of the planning assumptions (i.e. planners in the NHS for example) to gather their views as to how the planning assumptions were used in practice and which approaches they would find most helpful.
84. It was also felt by one interviewee that, although difficult, the definition of a reasonable worst case could benefit from further clarification. This could help to clarify the definition for those on the committee but would also help to

convey to planners what is meant by “reasonable”. For example, a definition could state something along the lines of “90% of scenarios are not going to be this bad”.

85. It was also questioned by some whether the reasonable worst case should be the upper bound of the anticipated range or whether they should be non contiguous, although no strong conclusions were conveyed.

ENGAGEMENT WITH CMO’S OFFICE

86. There was strong feeling that there was not enough engagement between CMO’s office and the SPI-M-O committee. It was highlighted that this would have been particularly beneficial in further understanding the CMO office’s collection of death data.
87. It was felt that it was understandable that the CMO’s office would want to quote some sort of estimate of the number of cases but greater engagement with CMO’s office might have meant that some of the issues with the case estimates (see page 12) might have been avoided.
88. There was also a significant delay in the committee getting access to the CMO’s report ahead of the CMO’s weekly briefings.
89. Some also highlighted that there was uncertainty as to whether the CMO’s office and the SPI-M-O committee had access to the same data sources, which could benefit from further investigation.
90. It was also suggested by one interviewee that the CMO’s office should have a representative on SAGE which is not currently the case and that this could help to improve engagement with the CMO’s office.

ACCESS TO AND USE OF DATA

91. In 2008 the DH influenza surveillance team carried out an exercise to specify surveillance requirements for an influenza epidemic response. The exercise involved consulting with relevant parties from the HPA (including those involved in specifying the RTM process), the NHS and stock management. The surveillance information requirements were signed-off by the Surveillance Pandemic Influenza Group (Surveillance PIG) and by the Pandemic Influenza Programme Board on 23rd July 2008.
92. During the epidemic, the DH surveillance team coordinated the sourcing of surveillance data (as specified during the surveillance exercise) and this included data from QSurveillance, NPFS, and other data collations (e.g. mortality information). Epidemiologists and DH analysts were able to access a lot of this data via the DH ImmForm website and in some cases directly from the surveillance schemes.

93. Paragraphs 94 to 110 set out the lessons learnt associated with surveillance data. It was generally felt that the surveillance specification should be revisited in light of these lessons.

Confidentiality and data usage agreements

94. Although a number of interviewees commented that they were used to data confidentiality agreements, it was felt by a few that the agreements could benefit from further clarity and adherence to them might be better enforced. One interviewee commented that they were unclear, at times, as to whether certain analysis was releasable to individuals outside of the committee.
95. It was generally felt that the various data usage agreements could be more transparent and in some cases DH could have been more forceful with some institutions about expectations regarding sharing of data.

Ethical approval

96. One interviewee commented that, for certain studies, ethical approval needed to be obtained at many levels which was particularly time consuming. For example, for one study, ethical approval needed to be obtained from the academic institution involved, the NHS at a national level and also from the NHS in each locality involved in the study. These delays led to considerable difficulties in achieving samples sizes required for the study. It was suggested, therefore, that it would be very beneficial for ethical approval to be pre-arranged for studies that are likely to be carried out during an epidemic.

ImmForm

97. It was commented by some that the ImmForm website was not ready at the start of the epidemic and that this led to some issues with the timeliness of data (however, the resulting issues were dealt with generally well). It was also felt by some that downloading data was initially difficult, though this became less so with time. One interviewee commented that data files could have been labelled more logically to assist with locating certain data.
98. It was suggested by one interviewee that since much data cleaning was being duplicated by the various modelling groups it might be more efficient for data cleaning to be carried out centrally (with transparency regarding how the data had been cleaned). This would effectively allow a central modeller's data warehouse to be created, possibly on the ImmForm website. It was also suggested that such a system could be used to store modelling papers as well as the data used in modelling for ease of reference.
99. One interviewee commented that the issues experienced with ImmForm should not be taken as a reflection on those that worked very hard to make the ImmForm system work and whom should be congratulated.

QSurveillance

100. Feedback indicated that getting more detailed data from QSurveillance could have been key, in particular, a breakdown of GP consultations by whether they were conducted face-to-face or over the phone and more detail as to the nature of GP consultations after NPFS was switched on (i.e. whether GPs were giving letters of authorisation for antivirals or directing patients to NPFS etc.).
101. It was also commented by some that it was difficult to understand the QSurveillance data usage agreements in terms of who owns the data, what one can do with the data and who to contact regarding the data.

Transition from case reporting to aggregate data

102. During pre-pandemic planning, there was some concern that there could be a significant gap between initial case reporting ceasing and a signal being identified through ILI measures. In practice, case reporting stopped in mid June with an effect in ILI showing around the end of June/ beginning of July and this transition also coincided with the move from the 'containment' to the 'mitigation' phase of the epidemic. There were some mixed views from interviewees as to whether this transition caused issues in terms of modelling. However, it was acknowledged that, outside of the SPI-M-O committee, the transition period and its impact on modelling output was not picked up as an issue as might have been expected.

Data from Northern Ireland, Scotland and Wales

103. It was commented by a few interviewees that the level of surveillance data that the committee had access to for Northern Ireland, Scotland and Wales was significantly less than that for England. For example the committee didn't have access to the Scottish situation report on a regular basis. As such it was unclear as to the extent of the committee's responsibility for modelling the epidemic in Wales, Scotland and Northern Ireland (see paragraph 12).

Hospitalisations and mortality tracking

104. It was commented by some that the hospitalisation and mortality tracking in England, was rather ad-hoc and the process could have been thought out better early on (and was not as good as the tracking systems used in Scotland).
105. It was, however, acknowledged by one interviewee that the process may necessarily need to be ad-hoc, since testing of all hospitalised cases for swine flu is unlikely to be a priority, and as such may be difficult to improve.
106. There was also feedback that the differences in hospitalisation policies and the recording of hospitalisation and deaths between the different UK countries led to considerable difficulties in comparing hospitalisation and mortality rates.

International data

107. It was felt by some that sharing of international data could be more formalised. Feedback indicated that getting access to anything more than the freely available international data and information over the internet was often problematic and seemed to be based upon informal links or contacts. It was suggested that there was a case for reviewing which countries would be comfortable with arranging memorandums of understanding in terms of data sharing ahead of a future epidemic.
108. It was also commented that ECDC had played some role in compiling international data during the epidemic but was unable to force the sharing of data since it is an advisory body.

Absenteeism data

109. The committee had a lot of pressure to report on the effects of the epidemic on absenteeism rates in the NHS and industry in general. In practice the committee (via the Cabinet Office) had considerable difficulties in accessing such data (very little absenteeism data was obtained from industry). It was felt that there needs to be a better system for the collection of absenteeism data (through the Department for Business, Innovation and Skills (BIS)) and that a strong message needs to be conveyed to industry and BIS that the collection of good absenteeism data is essential in order for the committee to make robust conclusions regarding the impact of an epidemic on absenteeism levels.

Data collection in the future

110. It was also highlighted that any lessons learnt in terms of data collection should bear in mind how data streams might change in the future.

BEHAVIOURAL DATA

111. A number of interviewees acknowledged that accurate attitudinal and behavioural data is key in interpreting flu surveillance data, in particular, the proportion with ILI who seek medical care and how this changes over time. It was felt that such aspects were (and still are to some extent) poorly understood. It was also commented that further understanding these aspects could be beneficial in helping DH to define its swine flu media strategy.
112. An online survey 'FluSurvey' was set up by LSHTM to obtain behavioural data, relating to the flu epidemic, directly from the public. Interviewees acknowledged the survey's limitations (e.g. that it was not random and relied

on participants opting-in) but felt that it provided valuable information, relatively cheaply, that couldn't be obtained anywhere else.

113. It was felt, however, that it would have been beneficial for FluSurvey to have been implemented sooner (in practice this wasn't possible due to resource issues).
114. It was also felt that the survey could also have significantly benefitted from better publicity since it had been difficult to get the survey onto the HPA website and was relatively hidden once it was there. Attempts were also made to get the survey onto the DH website but this was not achieved due to concerns as to whether the website was in-line with the DH communications strategy. It was suggested that the survey could benefit from being integrated into the HPA's overall surveillance strategy since this could alleviate the resource and publicity issues.
115. It was also commented that FluSurvey did demonstrate that the proportion seeking care changed over time (around the peak) and that this wasn't just due to NPFS since trends were seen in Wales where there was no flu service.

FF100 DATABASE

116. The FF100 database aimed to collect data and specimens from a limited number of the earliest laboratory confirmed cases and their contacts in order to enable early understanding of some of the key clinical, epidemiological and virological parameters of the swine influenza virus.
117. Some interviewees commented that the FF100 database was very important for estimating key epidemiological parameters at the outset of the epidemic. However it was felt by many that the protocol for the database was not ready at the outset of the epidemic. It was suggested that the need for an FF100 database was a predictable problem and as such the protocol should have been finalised ahead of time.
118. In practice, much of the protocol wasn't used and some suggested that it could have been whittled down to only capture the data that were most important, though it was acknowledged that this could be difficult since the database had multiple customers (i.e. clinicians and virologists as well as epidemiologists).
119. Feedback indicated that in practice the system was supplemented by other ad-hoc systems and was therefore fragmented in places. It was felt that, as such, only a small number of people had a sufficient handle on it.
120. Interviewees indicated that there are differing views as to whether the database should be primarily a public health or epidemiological endeavour. It was argued by some that since it is not possible to provide antivirals to contacts throughout the epidemic, it would be advantageous, from an

epidemiological perspective, to also not provide antivirals to contacts in the initial phase since this would lead to better epidemiological data.

121. It was also highlighted that labelling the initial phase of an epidemic as the “containment phase” lead to a focus on public health rather than epidemiology.
122. One interviewee felt that the process for accessing data from the FF100 early on in the epidemic was more complicated and time consuming than it should have been and suggested that DH could have been clearer regarding expectations with the HPA on sharing data with other parties.
123. It was acknowledge that the HPA has limited resources, however there were suggestions that the collection of data and specimens could have been better co-ordinated and resourced as those on the ground got quickly swamped. It was also felt that further thought should be put into whether resources should have been concentrated on expanding the number of cases in the FF100 or getting better data and serology for a smaller sample.
124. The database didn’t produce the serology data (from contacts of cases) that was intended and it was felt that this could have been key in helping to predict the peak of the epidemic. It was suggested that this could have been due to a lack of understanding on the ground of the epidemiological importance of collecting blood samples and also a reluctance from patients to provide samples.
125. More generally it was felt that the FF100 protocol should be further thought through in light of experiences from this epidemic

SEROEPIDEMIOLOGICAL STUDIES

126. The HPA conducted a number of seroepidemiological studies over the course of the epidemic. An initial study, presented to the SPI-M-O committee in August 2009, was aimed at understanding the baseline age-specific prevalence of immunity prior to the arrival of the swine influenza epidemic and a follow up study, presented to the committee in October 2009, was aimed at understanding the incidence of infection in England over time.
127. Some interviewees commented that the data from these studies was very useful, however, this contrasted with views from others that, whilst acknowledging that such studies are difficult to conduct, felt that the power of the studies was limited and therefore provided only limited insight. Some felt that, had the power of the studies been greater, this could have made a big difference to understanding the epidemiology of the virus. However, this contrasted with some comments from interviewees who felt unsure as to how much more insight practical serology studies could have provided (as opposed to reinforcing what was already known).
128. There was considerable uncertainty from a number of interviewees as to whether the serology studies were close to optimal (given the resource

constraints and the nature of the epidemic) or whether they could have benefited from more prioritisation and planning. If the studies were optimal then, it was argued by some, that there is a need to manage expectations ahead of time in terms of their power, or otherwise, to inform.

129. Some felt, however, that with greater prioritisation, a full serology protocol (with weekly or monthly sero-surveillance) could be set ahead of time that could be activated at the start of an epidemic, fully resourced through the HPA or other laboratories. It was also suggested that investment in serology could well pay for itself through understanding the epidemic better and the savings that this could lead to. Another interviewee suggested that there was potential to engage more with ministers to enable more in-depth serology studies if it was felt that this area wasn't exploited as much as it could have been and had the potential to make a great impact.
130. These comments contrasted with some views that SPI-M-O was overly involved with the serology studies and should leave the experts to conduct as best they can. It was also pointed out that, it was believed that the UK had serology data before most other countries, if not all.
131. One interviewee highlighted that, for certain epidemics, one would not expect any immunity in the population and as such serology studies would be futile. Therefore a serology protocol should not be automatically activated.
132. One interviewee commented that it would have been desirable to have had definitive analysis on the Eton outbreak. The outbreak of pandemic cases occurred in May with an initial analysis published in July but a full definitive report having not yet been published. It was unclear why this analysis had not yet been completed.
133. It was also commented that an opportunity was lost in not getting more schools involved in sero-testing of children. This could have provided valuable information for estimating attack rates since, unlike households, contacts in schools are (generally) not given prophylaxis during the initial phase of an epidemic.
134. It was also felt by one interviewee that the detail of the seroepidemiological studies could have been more accessible to the epidemiologists on the committee (i.e. the data rather than just the high-level reports).

SWABBING

135. Swabbing for pandemic H1N1 was conducted through the RCGP, RMN and NPFS swabbing schemes during the course of the epidemic. The positivity rates from these schemes were reported in the HPA situation report and used as key parameters in the HPA case estimates calculation.
136. It was felt by a number of interviewees that better understanding of the swabbing protocols for the different schemes would have been very beneficial. For example, it is still unclear what the sampling procedures were for each of the different schemes (e.g. a first through the door policy and

whether there were biases towards certain age groups etc.). It was also unclear whether protocols were kept constant throughout the epidemic

137. The sensitivity of the swabbing tests for the different schemes (and for those swabs taken by person or GP) is still unclear. Some felt that this is still an important research issue and could potentially be investigated by pairing together swabbing and serology data.
138. One interviewee questioned whether enough swabs had been taken during the course of the epidemic and whether the swabbing test could have been developed quicker at the beginning of the epidemic, which could have helped with early modelling attempts.

MANAGING EXPECTATIONS

139. It was felt strongly that there is not enough understanding of epidemiological modelling and analysis at senior official and ministerial levels and as such there is a strong need to manage expectations of what epidemiological modelling and analysis can and can't provide.

General epidemiological primer

140. It was strongly suggested by a number of interviewees that some sort of primer explaining the factors that limit epidemiological modelling and how these change over the course of an epidemic would be very beneficial. For example, this could include:
 - an explanation of the frailties of syndromic surveillance;
 - how surveillance changes over the course of an epidemic;
 - how and when serology data might become available and what the limitations of such data might be (see page 24);
 - how early analysis can be impacted by stochasticity (see paragraph 50);
 - how certain events and policy changes alter the consistency of data (e.g. the switching on of a flu service, antiviral and vaccination policy etc.);
 - how these limiting factors might differ depending upon the severity of the epidemic ;
 - how uncertainties in epidemiological model output does not necessarily reflect deficiencies in modelling as such but more often simply reflects a lack of data and that epidemiological modelling therefore requires robust data collection.
 - However, the modelling can explicitly indicate the range of uncertainty due to the uncertainty in disease parameters due to these data collection difficulties.

141. It was also commented by a couple of interviewees that the committee could benefit from some material for newcomers to the committee (in terms of the committee's function and the modelling it carries out) and therefore such a primer may be able to also fulfil that function.

Case estimates and Real Time Modelling (RTM)

142. It was also suggested that expectations may need to be managed in terms of case estimates (i.e. that producing case estimates may not be the best indicator of how an epidemic is progressing (see page 12)) and Real Time Modelling (RTM) (i.e. under some circumstances RTM might not be feasible, since it was suggested by some that the idea that RTM might not happen wasn't fully understood).

Planning assumptions

143. Managing expectations was also seen to be key in terms of the planning assumptions, see paragraph 71 for further details.

Two-way process

144. It was also identified by some that managing expectations is a two-way process and that the committee may benefit from further understanding the types of questions that ministers and senior officials need to ask during an epidemic.

APPENDIX A

LIST OF INTERVIEWEES

Name	Institution	Member or observer
Dr Peter Grove	Health Protection Analytical Team, Department of Health	Member
Professor Neil Ferguson	MRC Centre for Outbreak Analysis and Modelling, Imperial College London	Member
Dr Peter White	Centre for Infections (CfI), Health Protection Agency (HPA)	Member
Dr Steve Leach	Emergency Response Department, Health Protection Agency (HPA)	Member
Dr Ian Hall	Emergency Response Department, Health Protection Agency (HPA)	Member
Professor Matt Keeling	Mathematics Institute and Department of Biological Sciences, University of Warwick	Member
Professor John Edmunds	Infectious Disease Epidemiology Unit, London School of Hygiene and Tropical Medicine	Member
Professor Chris Robertson	Statistician, Health Protection Scotland	Observer

Dr Elaine Gadd	Pandemic Flu Team, Department of Health	Observer
Dr Peter Bennett	Health Protection Analytical Team, Department of Health	Observer
Dr Miles Elsdon (with comments from Judy Britton and James Dancy)	Civil Contingencies Team, Government Office for Science	Observer
Philippa Makepeace (with comments from Stephen Conroy)	Civil Contingencies Secretariat, Cabinet Office	Observer
Martin Hensher (with comments from Omar Idriss)	Flu Resilience Team, Department of Health	Observer
Guy Walker (with comments from James Haybyrne)	SPI-M-O Committee Secretariat, Department of Health	Observer

With comments from the following observers during a SPI-M meeting on 28th January 2010:

Luke Perera	Surveillance and Information Systems Manager, Pandemic Influenza Preparedness, Department of Health	Observer
Daniela DeAngelis	MRC Biostatistics Unit, Institute of Public Health, University of Cambridge	Observer

